E7: Introduction to Computer Programming
Midterm 2A

Monday, November 4

Full Name

Student ID Number

Instructions:

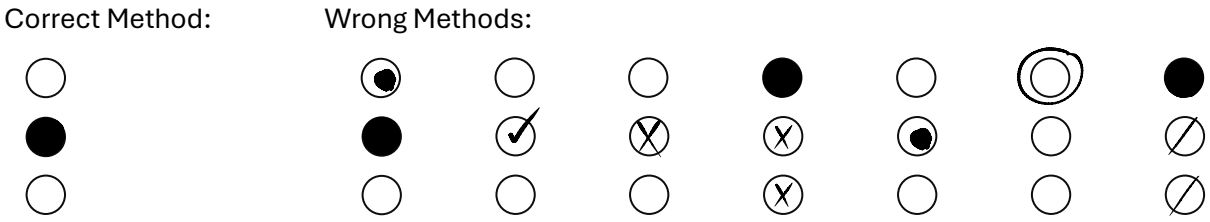
- You have 1 hour and 20 minutes to complete this exam.
- The exam is closed book, closed notes, closed computer, closed calculator.
- You may only use one 8.5" × 11" sheet of notes (both sides). No other resources are permitted.
- Mark your answers on the exam paper within the designated spaces. Responses written on scratch paper or outside of the specified answer spaces will not be graded.
- This exam is DOUBLE-SIDED. Make sure to read both sides of each page.
- The number of points associated with each part of each question is indicated.
- The exam is worth a total of 100 points.
- For questions with circular bubbles, you should select exactly *one* choice.
  - ◯ You may only chose one option!
  - ◯ Or this one, but not both!
  - ◯ Or this one, but no more than one!
- For questions with square boxes, you may select *multiple* choices.
  - ☐ You can select this one!
  - ☐ You can select this one too!
  - ☐ You can select one, two, or all three of these!
- Only include your FINAL answers in the designated boxes. Any additional text will count as part of your answer and may result in a reduced grade.

Good luck!

"We're rooting for you. You've got this!"

Question 1 (30 points)

There are 10 multiple choice questions. There is only **ONE** correct choice for each question. Multiple bubbles, incomplete bubbles, or stray marks will be marked incorrect.



1. (3 pt) Which error among the following is NOT caused by incorrectly written code?

   ○ Syntax error ○ Runtime error ● Round-off error ○ Logic error

   ○ None of the options (i.e., all are caused by incorrectly written code)

2. (3 pt) How do you create a vector from 1 to 10 (inclusive) with a step of 2 in MATLAB?

   ○ `linspace(1, 11, 2)` ○ `1:10:2` ○ `linspace(1, 10, 2)`

   ○ `1:11:2` ● `1:2:10` ○ `range(1, 10, 2)`

   ○ `linspace(1, 10, 5)` ○ `1:2:11` ○ `range(1, 11, 2)`

3. (3 pt) Assume that the MATLAB code below has been executed.

```
A = [1, 0; 2, 3; 5, 4];
B = [1, 0, 0; 0, 1, 0];
```

   Which of the following MATLAB expressions will NOT result in an error?

   ○ `A - B` ○ `A + B` ○ `A .* B` ● `A * B`

   ○ None of the options result in an error

4. (3 pt) In MATLAB, what is the purpose of the `end` keyword?

   ● End a block of code such as a loop or `if` statement

   ○ Stop the execution of the program ○ Terminate a loop early

   ○ Terminate a script ○ Close all open figures

5. (3 pt) How do you access the number 30 in the vector `v = [10 20 30 40 50]` in MATLAB?

   ○ `v(2)` ○ `v[3]` ● `v(3)` ○ `v[30]`

   ○ `v[2]` ○ `v(30)` ○ `v{3}` ○ `v{2}`

6. (3 pt) Regression is used to find

- ◯ the values at which a function is equal to zero
- ◯ a curve that exactly fits the data
- ◯ a curve that best fits the data by maximizing some measure of error
- ● a curve that best fits the data by minimizing some measure of error
- ◯ data points that are as close as possible to the actual data

7. (3 pt) What is the output of the following Python code?

```
x = 5
try:
    if x = 5:
        print(x + 1)
except SyntaxError:
    print(x + 2)
```

- ◯ `x + 1`
- ◯ `Nothing`
- ◯ `5`
- ◯ `6`
- ◯ `x`
- ● `Error`
- ◯ `7`
- ◯ `x + 2`

8. (3 pt) How do you insert a comment in MATLAB?

- ◯ `'This is a comment'`
- ◯ `# This is a comment`
- ● `% This is a comment`
- ◯ `// This is a comment`
- ◯ `/* This is a comment */`
- ◯ `"This is a comment"`

9. (3 pt) Which of the following is NOT a binary number?

- ◯ `01010101`
- ◯ `00000000`
- ◯ `11111111`
- ● `22222222`
- ◯ `01`
- ◯ `10000000`
- ◯ `10`
- ◯ `00000001`

10. (3 pt) Let `a` and `b` be two data structures of the same length. What does this code do?

```
import matplotlib.pyplot as plt
plt.plot(a, b)
```

- ◯ Creates a scatter plot of `b` (y-axis) versus `a` (x-axis)
- ◯ Plots the graph of `a` (y-axis) versus `b` (x-axis) in MATLAB
- ◯ Plots the graph of `b` (y-axis) versus `a` (x-axis) in MATLAB
- ● Plots the graph of `b` (y-axis) versus `a` (x-axis) in Python
- ◯ Plots the graph of `a` (y-axis) versus `b` (x-axis) in Python

Question 2 (7 points)

The following might be helpful for some of the next parts:

$2^2 = 4$, $2^3 = 8$, $2^4 = 16$, $2^5 = 32$, $2^6 = 64$, $2^7 = 128$, $2^8 = 256$.

$3^2 = 9$, $3^3 = 27$, $3^4 = 81$.

$4^2 = 16$, $4^3 = 64$, $4^4 = 256$.

$5^2 = 25$, $5^3 = 125$, $5^4 = 625$.

(a) (2 pt) What decimal number corresponds to the following signed binary integer: `10000011`

-3

(b) (2 pt) Convert the decimal number 10 into quaternary (base-4).

22

(c) (1 pt) In an 4-bit signed integer, what is the maximum positive value that can be represented? Provide your answer in 4-bit binary form, not the equivalent decimal number.

0111

(d) (2 pt) Why does this expression evaluate to `False` in Python? Select ALL that apply.

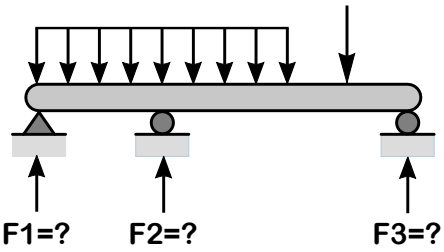
```
0.6 + 0.6 + 0.6 == 1.8
```

- [x] Accumulation of round-off errors
- [ ] Potential for underflow
- [ ] Python's floating-point representation introduces small inaccuracies
- [ ] Python represents decimal numbers precisely, so the expression should evaluate to `True`
- [ ] Python's addition operator has a bug when working with floating-point numbers
- [ ] All floating-point numbers in Python are approximations
- [ ] Potential for overflow
- [ ] Python uses single precision instead of double to represent floating-point numbers
- [x] Floating-point numbers are represented using a finite number of bits

Question 3 (15 points)

A structural element has three supports, and the reaction forces at each support are unknown and represented as: F1, F2, F3. Using equilibrium equations, you developed the following system of linear equations in matrix form $\mathbf{AF} = \mathbf{b}$:

$$\mathbf{AF} = \mathbf{b} \rightarrow \quad \begin{bmatrix} 1 & 1 & 1 \\ 0 & 3 & 9 \\ 3 & 0 & -6 \end{bmatrix} \begin{bmatrix} F1 \\ F2 \\ F3 \end{bmatrix} = \begin{bmatrix} 23 \\ 91.5 \\ -22.5 \end{bmatrix}$$



Note that understanding the diagram or any structural details is NOT necessary to solve this problem. The diagram is provided solely for illustrative purposes.

(a) (2 pt) Which system of linear equations corresponds to the matrix form shown above?

○ $\begin{cases} F1 + F2 + F3 = 23 \\ 3F2 + 9F3 = 91.5 \\ 3F1 - 6F2 = -22.5 \end{cases}$ ○ $\begin{cases} F1 + F2 + F3 = 23 \\ 3F1 + 9F2 = 91.5 \\ 3F1 - 6F3 = -22.5 \end{cases}$ ○ $\begin{cases} 3F1 = 23 \\ 12 = 91.5 \\ -3F3 = -22.5 \end{cases}$

● $\begin{cases} F1 + F2 + F3 = 23 \\ 3F2 + 9F3 = 91.5 \\ 3F1 - 6F3 = -22.5 \end{cases}$ ○ $\begin{cases} F1 + F2 + F3 = 23 \\ 3F1 + 9F2 = 91.5 \\ 3F1 - 6F2 = -22.5 \end{cases}$ ○ $\begin{cases} F1 + 3F3 = 23 \\ F1 + 3F2 = 91.5 \\ F1 + 9F2 - 6F3 = -22.5 \end{cases}$

(b) (2 pt) Using MATLAB syntax and *one line of code*, define matrix **A**.

```
A = [1, 1, 1; 0, 3, 9; 3, 0, -6]
% or [1 1 1; 0 3 9; 3 0 -6]
```

(c) (1 pt) Using MATLAB syntax and *one line of code*, define vector **b**.

```
b = [23; 91.5; -22.5]
% or [23, 91.5, -22.5]'
```

(d) (2 pt) You need to determine whether this system has a solution using the `rank()` function, which returns the rank of an input matrix. Given the defined matrix `A` and vector `b`, fill in the blanks of the code below to compute the required ranks, which should be saved as `rank1` and `rank2`. You only need to write the MATLAB code that computes the required ranks.

```
rank1 = rank(A)
% can switch the answers
```

```
rank2 = rank([A, b])
% can switch the answers
```

(e) (2 pt) After defining `rank1` and `rank2`, you obtained the results shown below. Which of the following is true regarding the system of linear equations?

```
>> rank1
rank1 =
        2

>> rank2
rank2 =
        2
```

- ◯ There are two solutions
- ◯ There is a unique solution
- ● There are infinite solutions
- ◯ There is a solution but can't determine if it is unique or infinite
- ◯ There are three solutions
- ◯ None of the options
- ◯ There is no solution

(f) (2 pt) After defining matrix `A` and vector `b` (parts b and c), you realized that you need to add a new equation to the system: $9F1 + 6F2 = 115.5$. Without using any functions and without directly redefining the entire matrices, use MATLAB syntax to update `A` and `b` to include this new equation.

```
A = [A; 9, 6, 0]
% or [A; 9 6 0]
% or [9 6 0; A]
```

```
b = [b; 115.5]
% or [115.5; b]
```

(g) (2 pt) After updating the matrices with the new equation, you need to solve for the unknown forces: F1, F2, F3. Complete the MATLAB code to solve for the unknown forces vector `F`.

```
F = A \ b
```

(h) (2 pt) Someone claims that the following vector is a solution to the system of linear equations:

$$sol = \begin{bmatrix} 4.5 \\ 8.5 \\ 10 \end{bmatrix}$$

Assume that `sol` is correctly defined. Which of the following MATLAB expressions could be used to check if vector `sol` is actually a solution to the system of linear equations?

- ◯ `sol * b == A`
- ◯ `sol * b = A`
- ◯ `b .* sol == A`
- ◯ `sol .* A == b`
- ◯ `sol .* b == A`
- ● `A * sol == b`
- ◯ `A .* sol == b`
- ◯ `A * sol = b`
- ◯ `A .* sol = b`
- ◯ `sol * A == b`
- ◯ `A .* b == sol`
- ◯ `A * b == sol`

Question 4 (8 points)

For all parts, assume that the following MATLAB function has been defined.

```
1  function c = fun(a, b)
2      c = 0;
3      if a > b
4          c = a - b;
5      end
6      if c < 10
7          return
8      else
9          c = 10;
10     end
11 end
```

What will be **displayed** when running the following expressions? If an error occurs, write "Error". If nothing is displayed, write "Nothing". If the evaluation would run forever, write "Forever".

(a) (2 pt) `fun(5, 2)`

```
3
```

(b) (2 pt) `fun(2, 5)`

```
0
```

(c) (2 pt) `fun(20, 5)`

```
10
```

(d) (2 pt) Which of the following will NOT change what is displayed when calling `fun(20, 5)`? Select ALL that apply. Line numbers are provided on the left side of the function.

- ■ Changing line 2 from `c = 0;` to `c = 20;`
- ■ Removing all indentation
- ☐ Changing line 9 from `c = 10;` to `c = 20;`
- ☐ Deleting lines 5, 6, and 7
- ☐ Deleting lines 8 and 9 (the `else` block)
- ☐ Removing the semicolons `;` at the end of lines 2, 4, and 9
- ☐ Deleting all `end` keywords

Question 5 (10 points)

Consider a file named `'file.txt'` that contains the contents to the right. This file is located in the same directory and is accessible to your code.

```
5 9 4
3 1 7
```

(a) (2 pt) What does the expression `type(a)` return after running the following code?

```
with open('file.txt') as f:
    a = f.read()
```

- ● `str`
- ○ `float`
- ○ `ndarray`
- ○ `list`
- ○ `tuple`
- ○ `int`

(b) (2 pt) What does the expression `type(b)` return after running the following code?

```
with open('file.txt') as f:
    b = f.readlines()
```

- ○ `str`
- ○ `tuple`
- ○ `int`
- ○ `float`
- ○ `ndarray`
- ● `list`

(c) (2 pt) Which option below shows the contents of `'file.txt'` after running this code?

```
with open('file.txt', 'w') as f:
    f.write('1 6 9')
```

- ○
```
5 9 4
3 1 71 6 9
```
- ○
```
5 9 4
3 1 7 1 6 9
```
- ○
```
5 9 4
3 1 7
1 6 9
```
- ●
```
1 6 9
```

(d) (2 pt) When creating scatter plots using `matplotlib`, what is(are) some advantages of using `plt.scatter()` over `plt.plot()`? Select ALL that apply.

- ☐ There are no differences; `plt.scatter()` and `plt.plot()` behave the same
- ☐ `plt.scatter()` automatically generates gridlines for better visualization
- ☐ `plt.scatter()` automatically connects data points with lines
- ■ `plt.scatter()` allows you to customize the size of individual points
- ■ `plt.scatter()` allows you to customize the color of individual points

(e) (2 pt) When creating 3D surface plots in `matplotlib`, which function is typically used to generate the grid of `X` and `Y` coordinates needed for the plot? Note that `np` is the abbreviated form for `numpy`, while `plt` refers to `matplotlib.pyplot`.

- ○ None of the options
- ○ `range()`
- ○ `plt.plot_surface()`
- ○ `np.linspace()`
- ○ `plt.grid()`
- ○ `np.arange()`
- ● `np.meshgrid()`
- ○ `plt.plot()`
- ○ `plt.plot3D()`

Question 6 (15 points)

In our course, the midterm grading policy is as follows:

- The lower midterm grade remains unchanged and contributes a weight of 1 (total of 100 points).
- The higher midterm grade is doubled and contributes a weight of 2 (total of 200 points).
- If both midterm grades are equal, the weight of 2 is assigned to midterm 1.
- The total midterm grade for each student is then calculated by summing the adjusted grades, resulting in a maximum score of 300 points.

You need to write a MATLAB function named `grade(m1, m2)` that takes two column vectors, `m1` and `m2`, of equal length. Note that the data are organized such that each element in `m1` corresponds to the midterm 1 grade of a student, and the element in the same position in `m2` corresponds to the midterm 2 grade of that same student. Also, assume that `m1` and `m2` are complete with no missing values. The function should perform the following tasks:

1. Create two new vectors, `w_m1` and `w_m2`, which assign a weight of either 1 or 2 to each student's midterm 1 and midterm 2 grades, respectively, according to the grading policy.
2. Calculate and return a vector called `total`, which contains the total score for each student by applying the weights to their respective midterm grades.

For example, with midterm grades: `m1 = [70; 90; 80; 90]` and `m2 = [80; 80; 80; 100]`

In this case, the weights would be: `w_m1 = [1; 2; 2; 1]` and `w_m2 = [2; 1; 1; 2]`

Finally, the function should return: `total = [230; 260; 240; 290]`

Anything written on this page will NOT be graded.

```
function ______(a)______
    % Initialize weight vectors w_m1 and w_m2 of the same size as m1 and m2
    w_m1 = ones(size(m1));
    w_m2 = ones(size(m2));

    % Update weight vectors w_m1 and w_m2
    ______(b1)______;
    ______(b2)______;

    % Calculate total score for each student
    total = ______(c)______;
end
```

You will implement the function `grade(m1, m2)` by filling in the blanks. You may only write one statement on each line. You may not use ; or , to place two statements on the same line.

Use MATLAB syntax for all parts on this question.

(a) (3 pt) Which of these could fill in blank (a). Select ALL that apply.

- ☐ `grade(m1, m2) = total`
- ☐ `grade(m1, m2)`
- ☐ `grade(m1, m2) = [total]`
- ☐ `def grade(m1, m2):`
- ■ `total = grade(m1, m2)`
- ■ `[total] = grade(m1, m2)`

(b) (3 pt) Fill in blanks (b1) and (b2) to update `w_m1` and `w_m2`, respectively. Do NOT redefine the vectors or use ANY functions. Answers that redefine `w_m1` or `w_m2` or use any functions will receive 0 points. Keep both `w_m1` and `w_m2` as column vectors.

Fill in blank (b1) to update `w_m1`.

```
w_m1(m1 >= m2) = 2
```

Fill in blank (b2) to update `w_m2`.

```
w_m2(m1 < m2) = 2
```

(c) (3 pt) Which of these could fill in blank (c)?

- ◯ `m1 x w_m1 + m2 x w_m2`
- ◯ `w_m1 + w_m2`
- ◯ `m1 .* 1 + m2 .* 2`
- ◯ `m1 * w_m1 + m2 * w_m2`
- ◯ `m1 . w_m1 + m2 . w_m2`
- ● `m1 .* w_m1 + m2 .* w_m2`
- ◯ `m1 + m2`
- ◯ `m1 ** w_m1 + m2 ** w_m2`
- ◯ `m1 + w_m1 + m2 + w_m2`

(d) (4 pt) Redefine function `grade(m1, m2)` in a **single** line of code that entirely replaces both the function definition and its body. Directly calculate the total scores using the midterm grades, without defining separate intermediate variables for the weights. For this part, you must use the following built-in MATLAB functions, where `A` and `B` are two arrays:

- `min(A, B)`: returns an array of the smaller values between corresponding elements of `A` and `B`
- `max(A, B)`: returns an array of the larger values between corresponding elements of `A` and `B`

```
grade = @(m1, m2) min(m1, m2) + 2 * max(m1, m2);
```

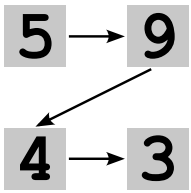
(e) (2 pt) Repeat part (d) without using ANY functions. This part is only worth 2 points, and may be challenging. Finish the rest of the exam first! No partial credit will be awarded.

```
grade = @(m1, m2) m1 + m2 + (m1 < m2) .* m2 + (m1 >= m2) .* m1;
% Other solutions very likely
```

Question 7 (15 points)

You need to write a MATLAB function `Reshape(vec, rows, cols)` which takes a vector `vec` and two positive integers `rows, cols`. The function should reshape `vec` into a matrix with a number of rows equal to `rows` and a number of columns equal to `cols` by rearranging existing elements, and it should return the matrix. All the elements from the input vector must fit into the new matrix, and the order of the elements in the input vector should be preserved. Specifically, the elements should fill each row from left to right before moving to the next row. See the schematic below.

```
>> Reshape([5 9 4 3], 2, 2)
ans =
     5    9
     4    3
```



Since all elements in `vec` should appear in the new matrix, not all values of `rows` and `cols` are compatible with `vec`. If the inputs are incompatible, the function should display an error message.

Test Cases (MATLAB Syntax):

```
>> Reshape([5 9 4 3 1 7], 3, 2)
ans =
     5   9
     4   3
     1   7
```

```
>> Reshape([5 9 4 3 1 7], 2, 3)
ans =
     5   9   4
     3   1   7

>> Reshape([5 9 4 3], 3, 2)
Error using Reshape
```

There are different ways to implement this function. The first method will require you to assign values from the input vector into the new matrix named `result` one value at a time.

```
function result = Reshape(vec, rows, cols)
    if numel(vec) ______(a)______
        error("Number of elements not compatible with rows and cols.")
    end
    result = ______(b)______; % initialize new matrix with the correct size
    index = 1;                % keep track of the current element in vec
    for i = ______(c)______
        for j = ______(d)______
            ______(e)______;
            index = ______(f)______;
        end
    end
end
```

You will implement the function `Reshape()` by filling in the blanks. You may only write one statement on each line. You may not use ; or , to place two statements on the same line. You may not use ANY functions, unless given as multiple choices. You should use MATLAB syntax.

(a) (2 pt) Fill in blank (a). Note that `numel(A)` returns the number of elements in array `A`.

```
if numel(vec) ~= rows * cols
```

(b) (2 pt) Which of the following options could fill in blank (b)? Select ALL that apply.

- ■ `ones(rows, cols)`
- ☐ `ones(cols, rows)`
- ☐ `[rows; cols]`
- ☐ `[cols, rows]`
- ☐ `zeros(cols, rows)`
- ■ `zeros(rows, cols)`
- ☐ `[rows, cols]`
- ☐ `[cols; rows]`

(c) (2 pt) Fill in blank (c).

```
for i = 1:rows
% or 1:1:rows
```

(d) (2 pt) Fill in blank (d).

```
for j = 1:cols
% or 1:1:cols
```

(e) (3 pt) Fill in blank (e).

```
result(i, j) = vec(index)
```

(f) (1 pt) Fill in blank (f).

```
index = index + 1
```

STOP! READ THE TWO LINES BELOW CAREFULLY!

The last part of the exam is only worth 3 points, and some might find it challenging. Finish the rest of the exam first!

Next, you will implement the same function `Reshape(vec, rows, cols)` but using two different methods. This second method will require you to assign each column of `result` (one entire column at a time) instead of assigning one value at a time. Assume that the inputs are compatible, and therefore, *you do not need* to check them for the error.

```
function result = Reshape(vec, rows, cols)
    % initialize new matrix with the correct size
    result = % (same answer as part b)
    for i = ______(g)______
        ______(h)______;
    end
end
```

You will implement the function `Reshape()` by filling in the blanks. You may only write one statement on each line. You may not use ; or , to place two statements on the same line. You may not use ANY functions.

(g) (0.5 pt) Fill in blank (g).

```
for i = 1:cols
% or 1:1:cols
```

(h) (1 pt) Fill in blank (h). No partial credit will be awarded.

```
result(:, i) = vec(i:cols:end)
```

This third method will require you to assign each row of `result` (one entire row at a time) instead of assigning one value at a time. Assume that the inputs are compatible, and therefore, *you do not need* to check them for the error.

```
function result = Reshape(vec, rows, cols)
    % initialize new matrix with the correct size
    result = % (same answer as part b)
    for i = ______(i)______
        ______(j)______;
    end
end
```

You will implement the function `Reshape()` by filling in the blanks. You may only write one statement on each line. You may not use ; or , to place two statements on the same line. You may not use ANY functions.

(i) (0.5 pt) Fill in blank (i).

```
for i = 1:rows
% or 1:1:rows
```

(j) (1 pt) Fill in blank (j). No partial credit will be awarded.

```
result(i, :) = vec(1+(i-1)*cols:i*cols)
```